Friedrich-Alexander-Universität
Technische Fakultät

The Impact of Agentic AI on Science

Invited Talk • PULS Retreat 2026

Andreas Maier

Friedrich-Alexander-Universität Erlangen-Nürnberg 11 October 2026 • Hotel Kolovare, Zadar

Outline

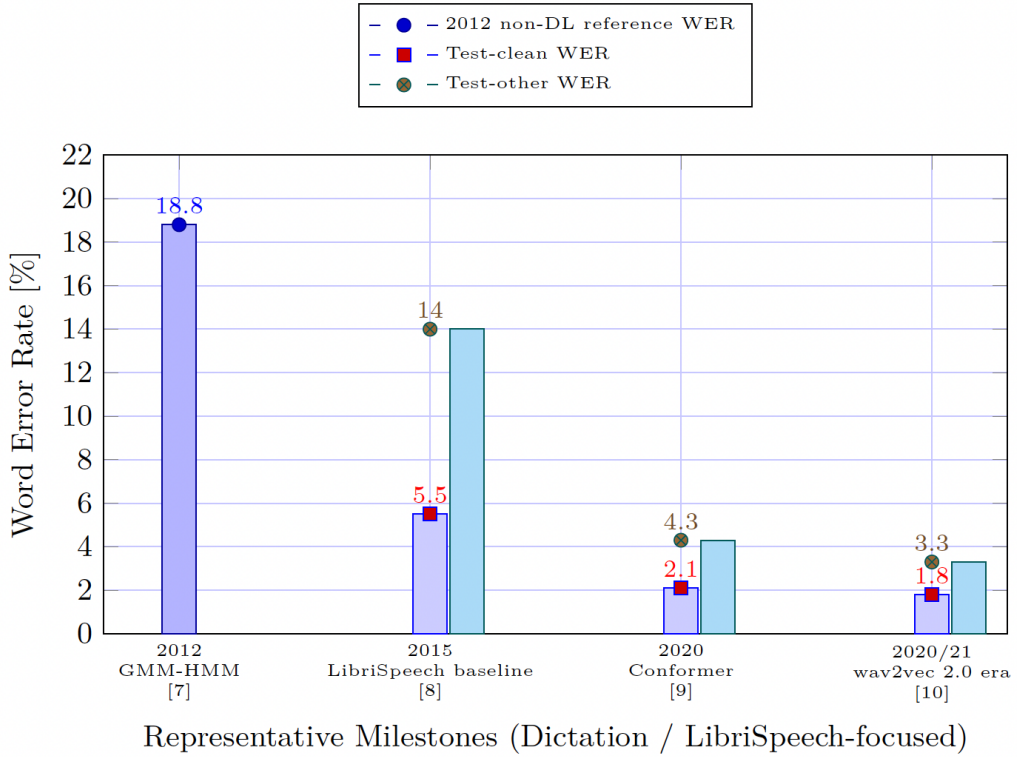
- What is deep learning?
- What is agentic AI?
- What does it mean for science?

What is deep learning?

Deep learning

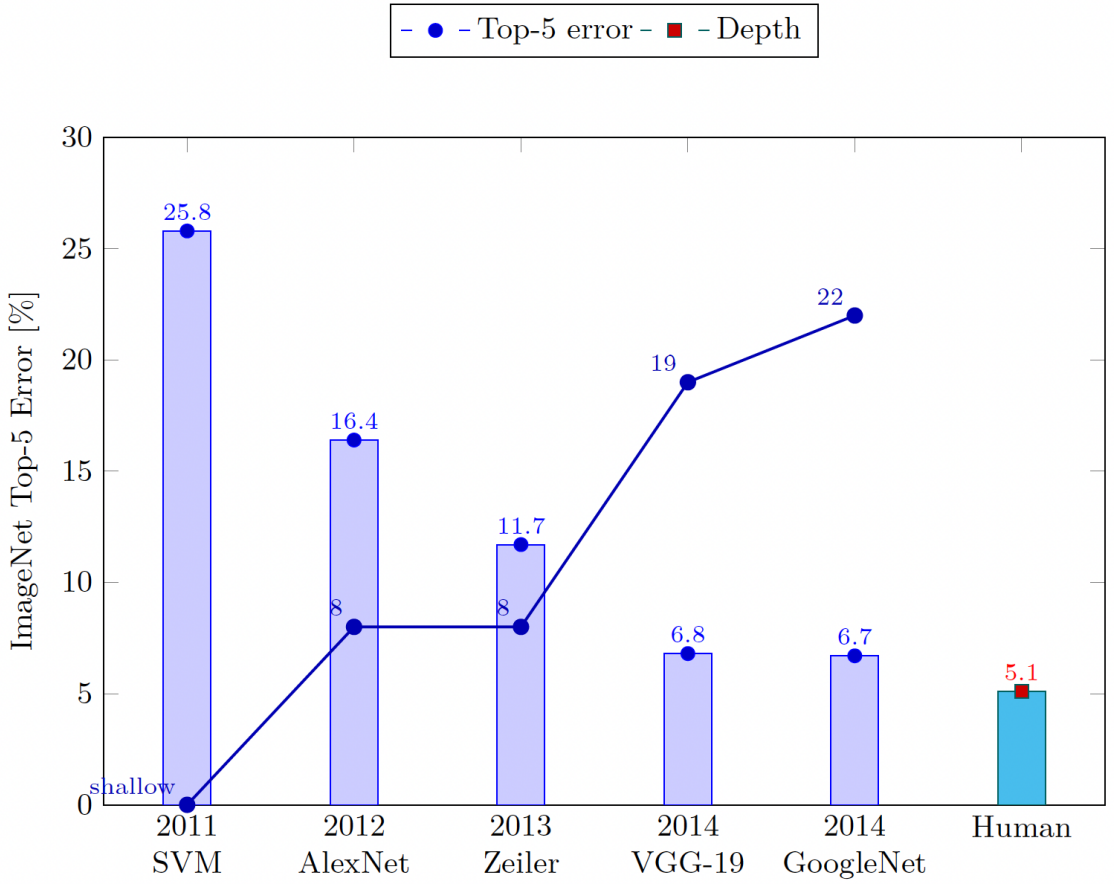
Speech

- Revolutionising pattern recognition and AI for more than 10 years.



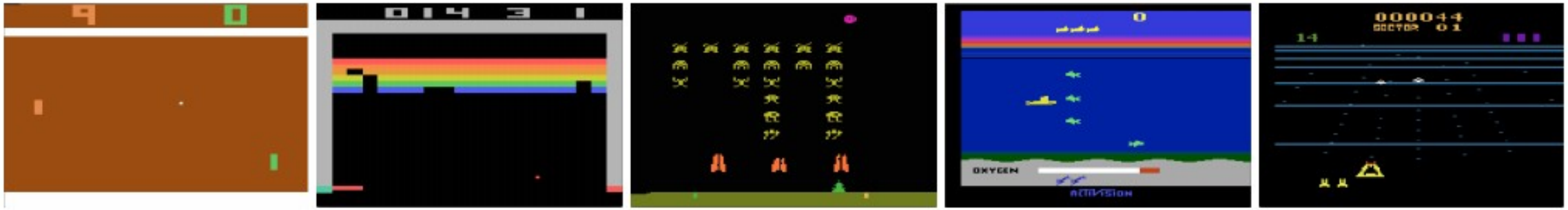
Deep learning

Image classification



Deep learning

Learning to play



[Mnih 2013]

- Deep Q-Networks learn Atari games from raw pixels alone.
- One agent, many games – super-human on several titles.

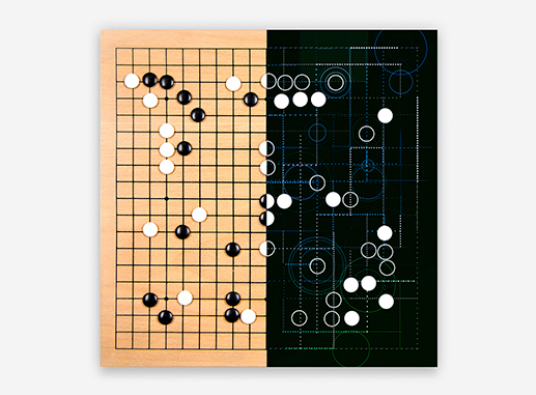
Deep learning

Learning to play



[Mnih 2013]

- Deep Q-Networks learn Atari games from raw pixels alone.
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AlphaGo (DeepMind, 2016)

- Self-play + Monte-Carlo tree search.
- Defeats world champion Lee Sedol 4–1.

Generative AI

Source image



Generative AI

Image description



“An athletically built man with a serious expression stands in a well-lit room with a wooden ceiling. He is wearing a blue t-shirt with the phrase ‘SAVE THE CHUBBY UNICORNS’ and an image of a rhinoceros. The man is taking a selfie with a modern smartphone, capturing his reflection in a mirror. In the background, one can see gym equipment including dumbbells and a weight rack. There is also a hanging chandelier with a unique design, illuminating the room.”

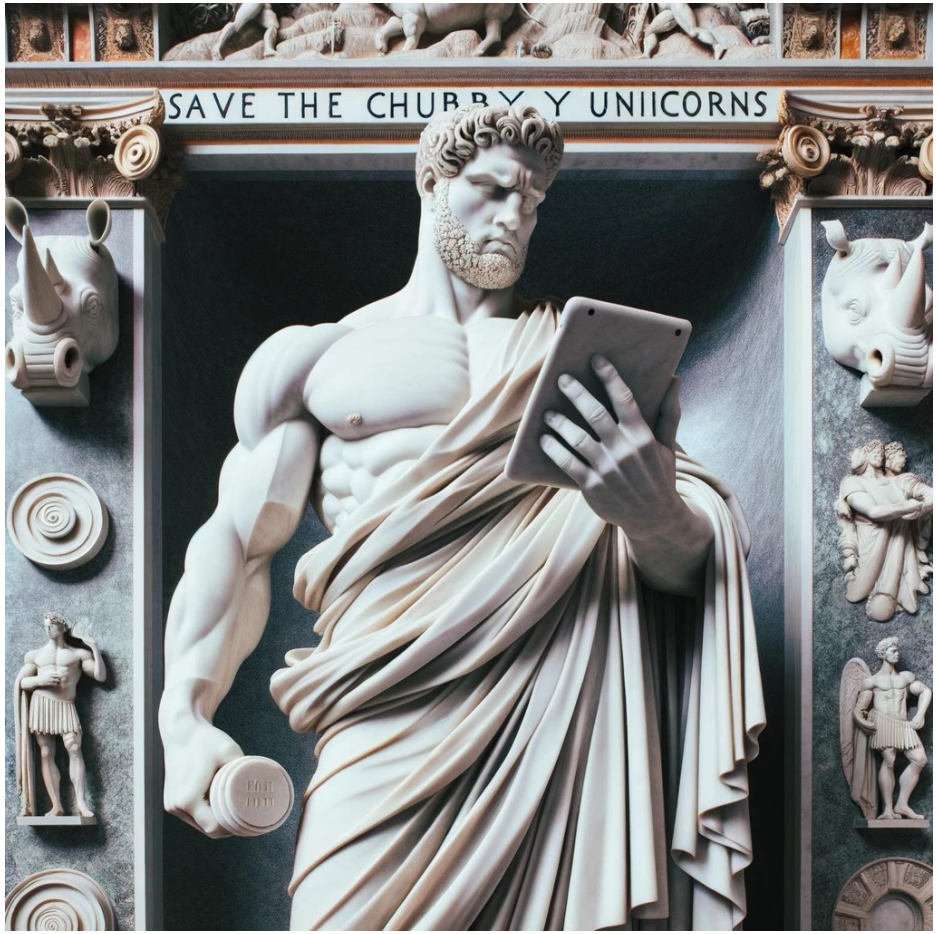
Generative AI

Generated image



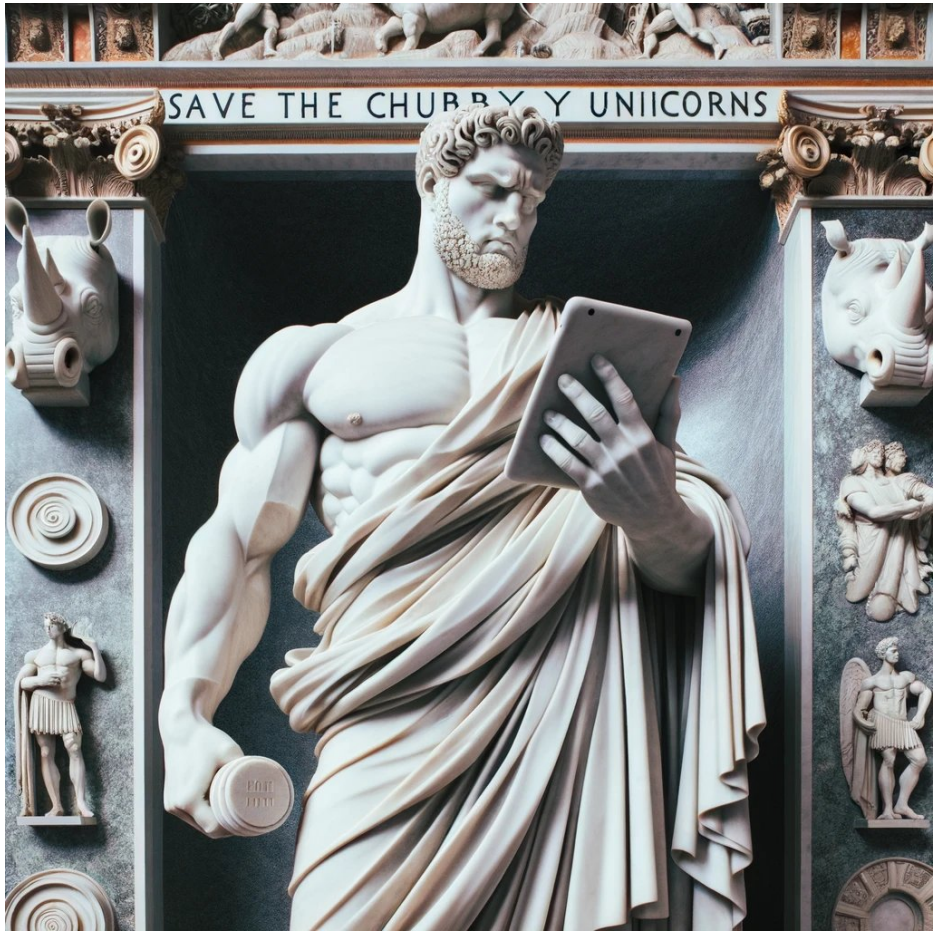
Generative AI

Style variations



Generative AI

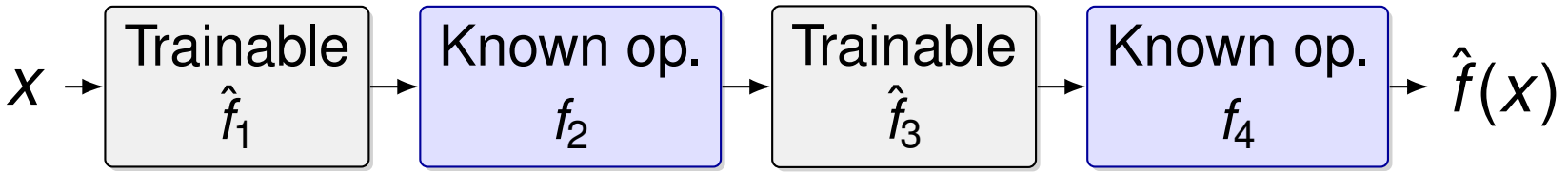
Style variations





Deep learning in medicine

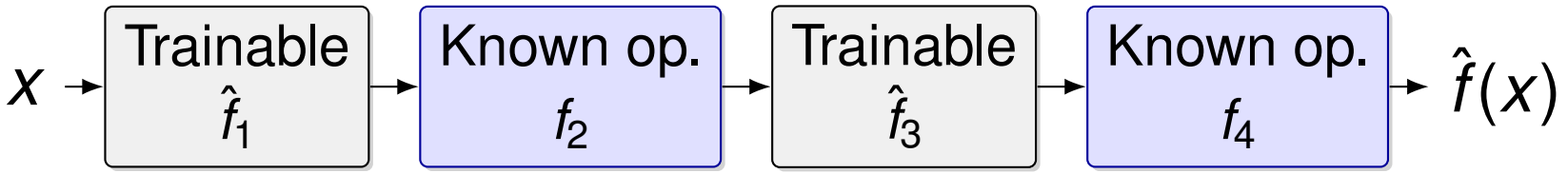
Known operators



Mix *trainable* layers with deterministic operators (Fourier transform, gradient descent, projection, . . .).

Deep learning in medicine

Known operators



Mix *trainable* layers with deterministic operators (Fourier transform, gradient descent, projection, ...).

Theorem – known operators reduce the error bound

For $f = f_n \circ \cdots \circ f_1$ approximated by $\hat{f} = \hat{f}_n \circ \cdots \circ \hat{f}_1$ with per-layer errors ε_i and Lipschitz constants $\hat{L}_j$,

$$\varepsilon(\hat{f}) \leq \sum_{i=1}^{n} \left(\prod_{j=i+1}^{n} \hat{L}_j \right) \varepsilon_i .$$

Replacing $\hat{f}_k$ with the *known* f_k forces $\varepsilon_k = 0$ and strictly tightens this bound.

Maier et al., *Nature Machine Intelligence*, 2019.

Deep learning in medicine

Recognition





Deutscher Zukunftspreis – low-field MRI made possible by deep learning.

What is agentic AI?

What is agentic AI?

A simple question

“How do you drive a car?”

Thought for a few seconds. . .

- A classic prompt to a large language model.
- After a short “thinking pause” the answer arrives as long-form running text.
- No tools, no actions – the model just *describes* how to drive.

What is agentic AI?

Long text answer of an LLM

To drive a car, follow these steps:

- **Get in and adjust:** fasten your **seat belt**, set the **seat** and **mirrors** so you can see the road clearly.
- **Start the engine:** press the **brake pedal**, turn the key or push the start button. Make sure the gear selector is in **Park (P)** or **Neutral (N)**.
- **Engage a gear:** for an automatic, move the lever to **Drive (D)**; for a manual, press the **clutch** and select **1st gear**.
- **Release the parking brake** and slowly lift your foot off the brake. Gently press the **accelerator** to move forward.
- **Steer** smoothly with both hands and **check mirrors** regularly. Keep a safe distance from the vehicle in front.
- **To stop:** ease off the accelerator, press the brake, and bring the car to a complete stop. Shift to **Park** and apply the parking brake before turning off the engine.

Always observe traffic rules and adapt your driving to road and weather.

$\Rightarrow$ **Lots of words, but no *action*.**

What is agentic AI?

From language model to agent



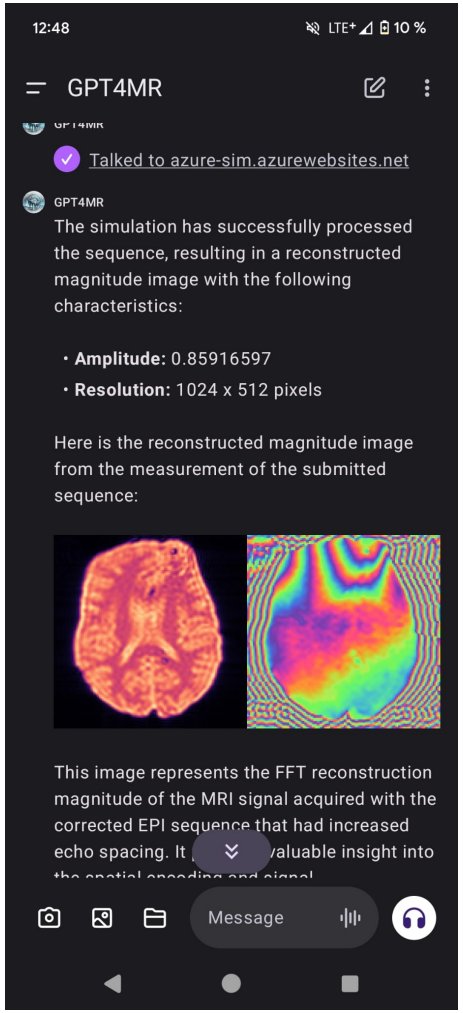
Moritz Zaiss

What is agentic AI?

From language model to agent



Moritz Zaiss



Zaiss et al., *GPT4MR*, 2024.

What is agentic AI?

Task for the models

"Code a spin echo EPI (64×64, fov = (0.2, 0.2, 0.08) m, TE = 150 ms)!"

- **Spin-echo EPI** – an advanced MRI sequence used for diffusion and functional imaging.
- Three models receive the *same* natural-language instruction and must produce runnable Python code.
- Contestants:

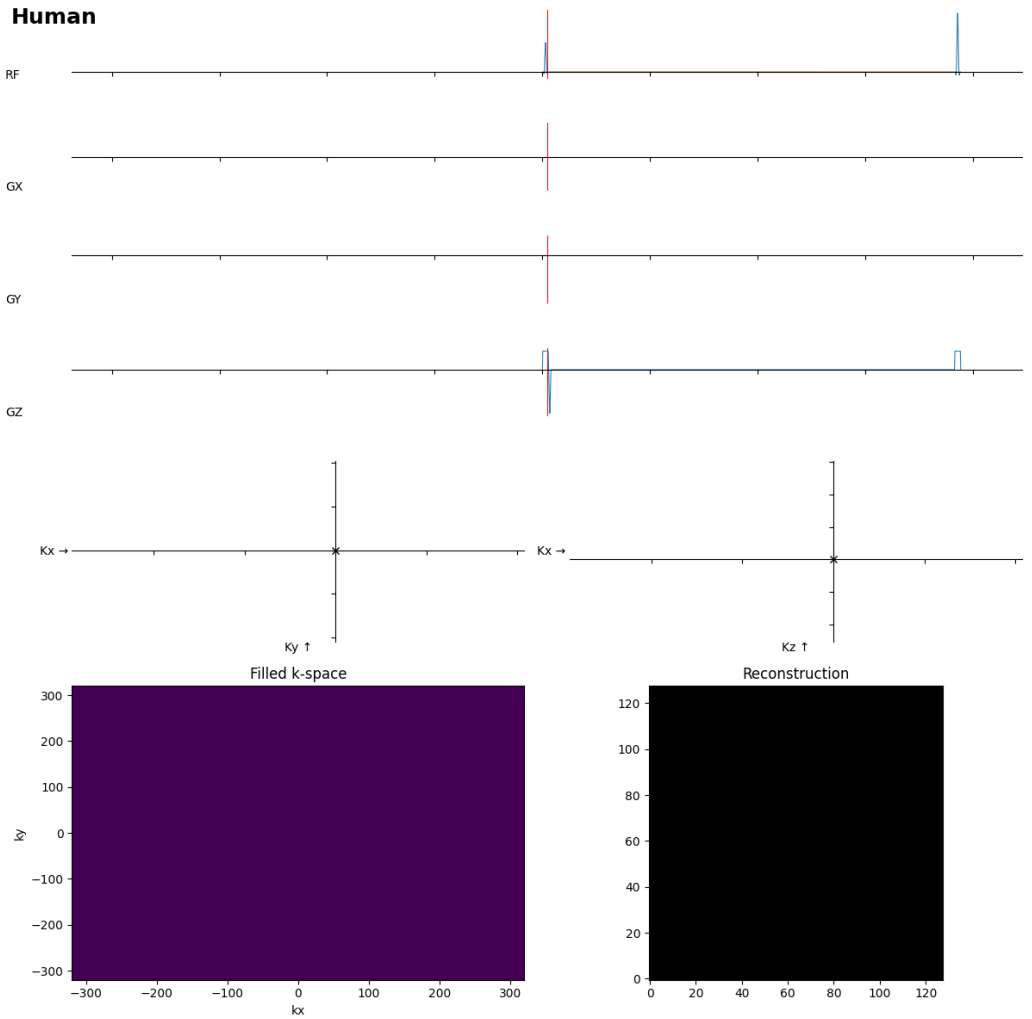
Human
reference

GPT-4o
OpenAI

GROK-3-think
xAI

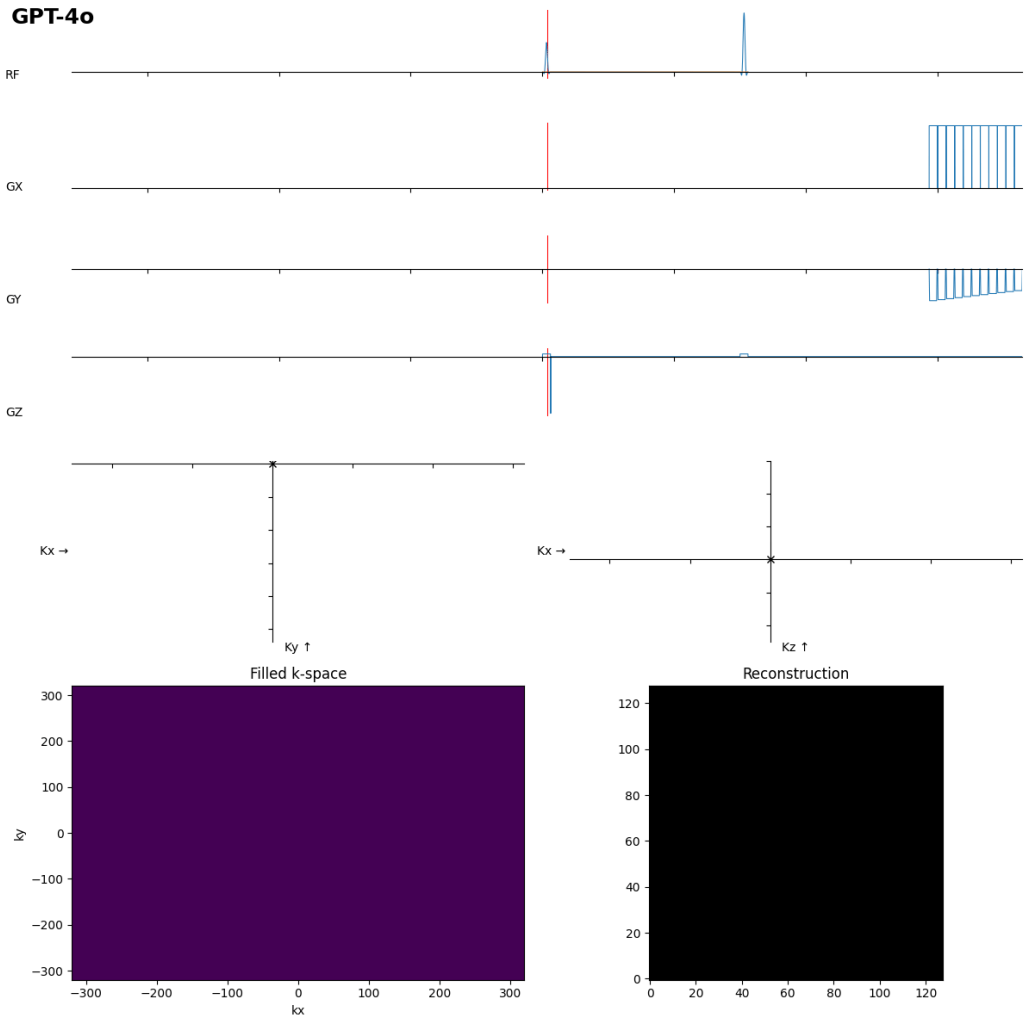
What is agentic AI?

Human baseline



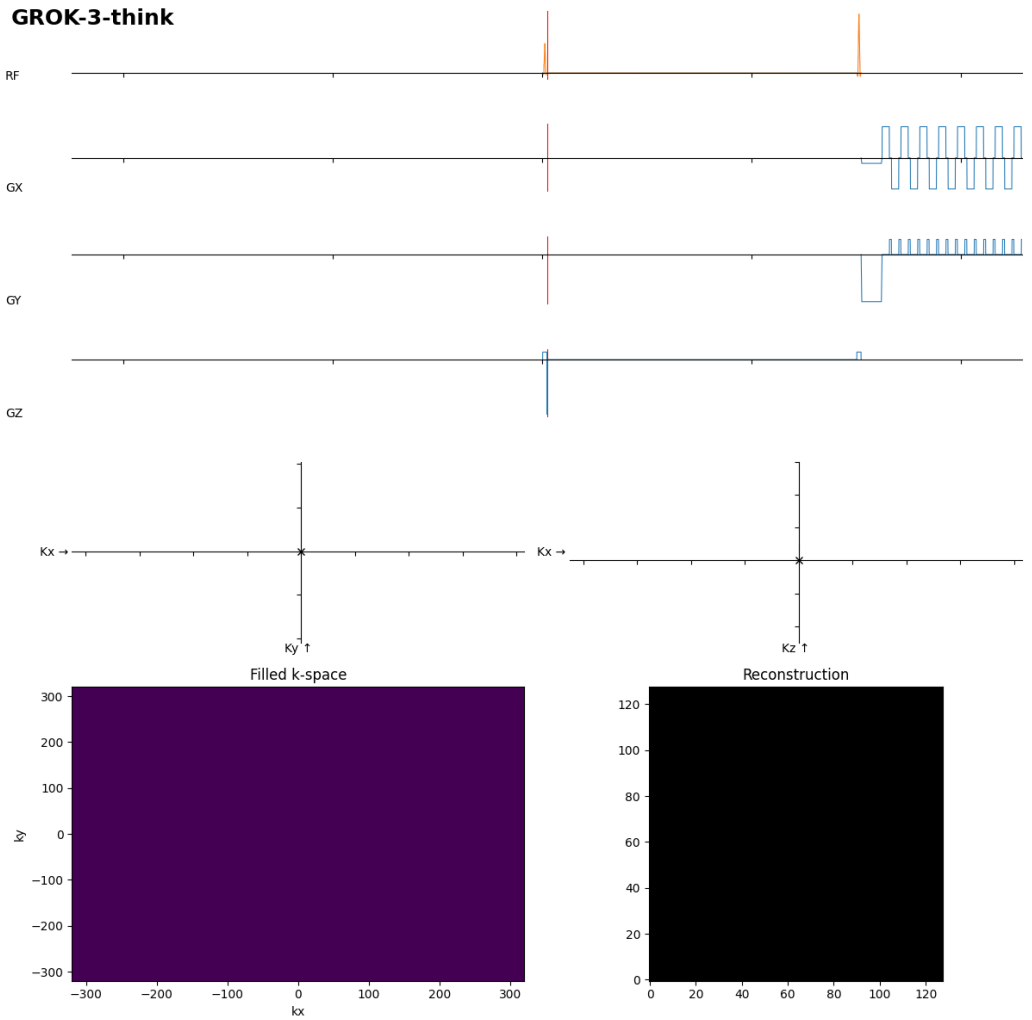
What is agentic AI?

GPT-4o attempt



What is agentic AI?

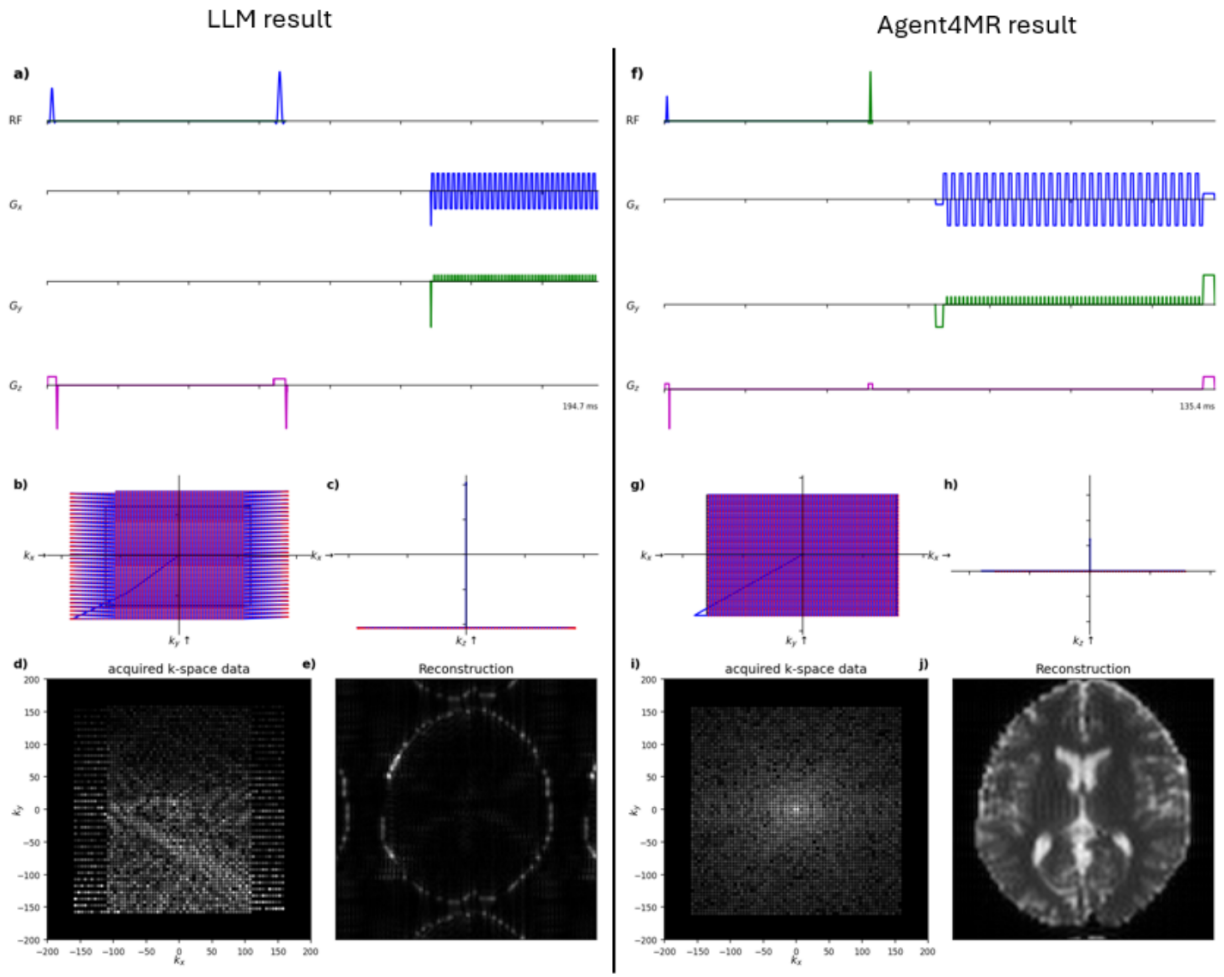
GROK-3-think attempt



How about real agentic AI?

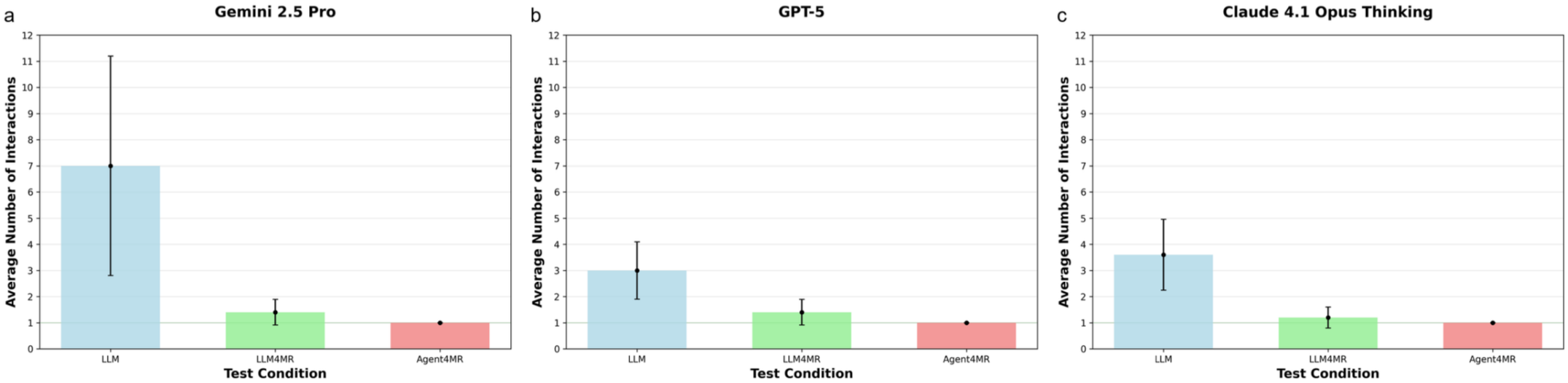
What is agentic AI?

LLM vs. Agent4MR



What is agentic AI?

Interactions per model



What is agentic AI?

Cost comparison

Table 1: User interactions N, time, token usage, and estimated cost until a satisfactory sequence. Human baseline: recently trained EPI-naive student, 2.5 h, 5 interactions.

Configuration Agent4MR	N	Time [min]	Tokens	Cost [€] [≈]
Gemini 2.5 Pro	1	4.8	87k	0.12
GPT-5	1	15	520k	0.48
Claude 4.1 Opus	1	10.2	1162k	10.00
Human	5	150	—	37.50

What is agentic AI?

Iterative progress

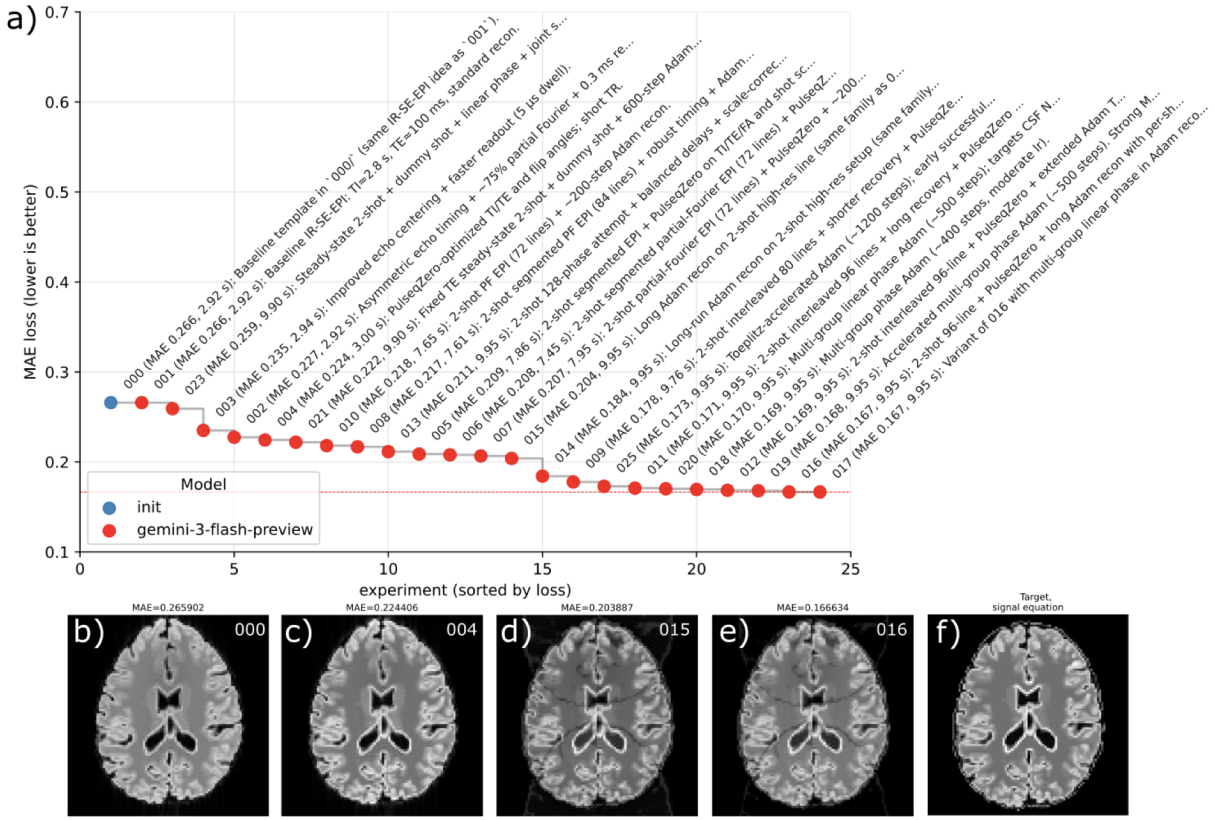


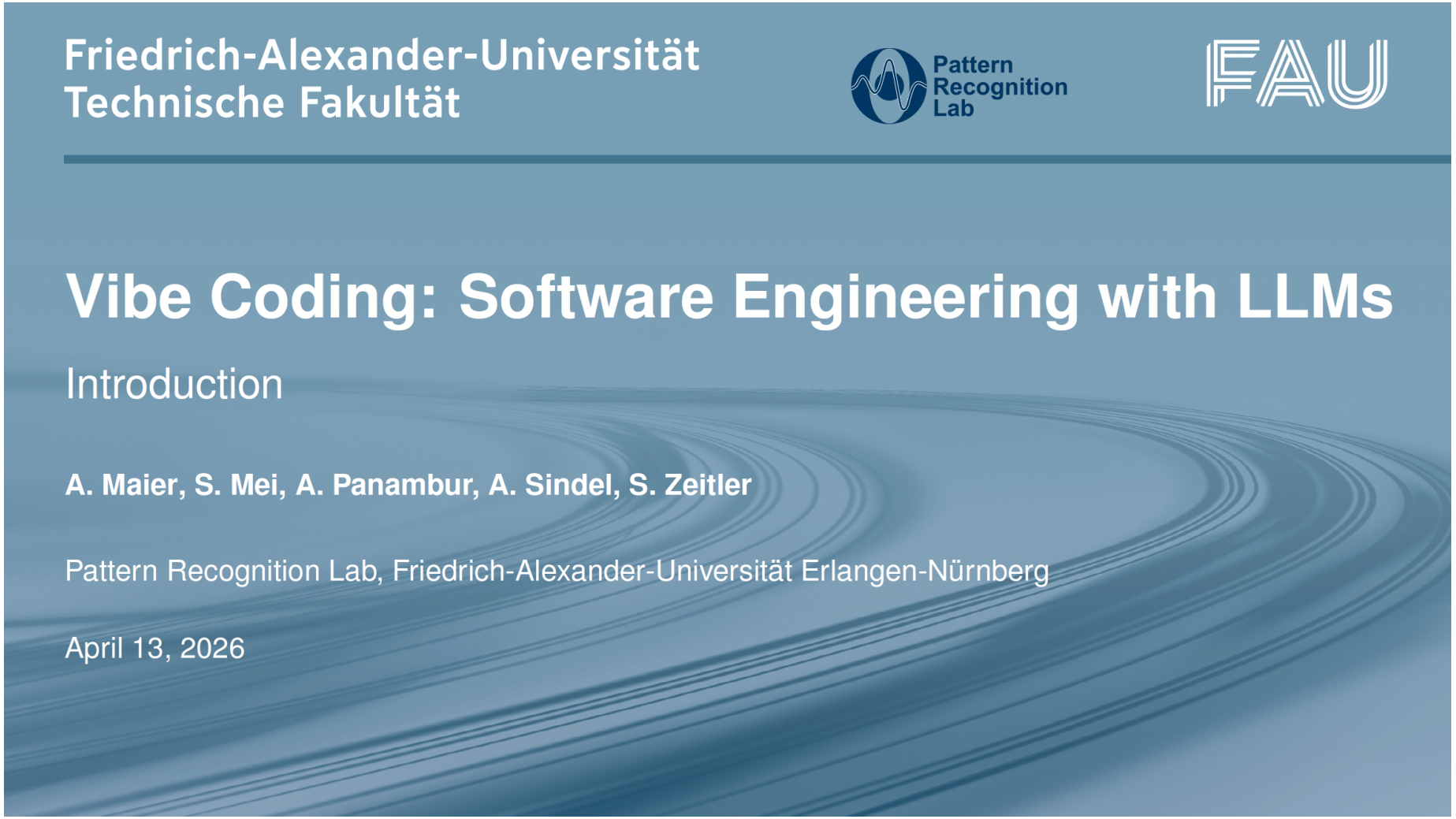
Figure 5: IR-SE-EPI MAE leaderboard progress. Autonomous agents iteratively improved the IR-SE-EPI pipeline (sequence parameters, reconstruction, post-processing). Experiments are ranked by MAE loss (worst to best). The staircase shows improvement from baseline (∼0.2659) to best (∼0.1666) through multi-window filtering, multi-shot introduction, phase correction, and differentiable TI/TE optimization.

Table 2: Normalized Token Usage and API Costs in EUR (n=25 events per model)

Gemini Model	Total Tokens [Mio]	Tokens per Exp. [Mio]	Cost [€]
2.5 Flash	25.456	1.018	2.43
3 Flash	136.340	5.453	12.85
3.1 Pro	72.039	2.881	59.72
Total	**233.836**	**9.353**	**75.00**

What is agentic AI?

Vibe coding



What is agentic AI?

Advanced AI-assisted techniques

Advanced AI-Assisted Techniques

Claude Skills [3]:

- Shareable workflow automation (YAML + Markdown)
- Team-wide best practices as code
- Deployment, testing, code review
- Available at `github.com/anthropics/skills`

Model Context Protocol (MCP):

- Standardized tool integration
- Connect AI to databases, APIs, files
- Growing ecosystem of plugins

Other Advanced Patterns:

- **Agentic workflows**: Multi-step autonomous tasks
- **Tool-augmented coding**: AI uses external tools
- **Vision support**: Screenshot-driven debugging

Key Benefit

These techniques enable AI to perform complex, multi-step workflows beyond simple code generation.

to a programmer interface for an agent.

Pattern Recognition Lab **A. Maier** et al. Vibe Coding: Software Engineering with LLMs – Introduction December 2, 2025 11/27

What is agentic AI?

Markdown-driven quality gates

Geek Box 13.3: Markdown-driven quality gates: when agents run the production checklist

A key insight from agentic software engineering is that quality criteria, which used to require human inspection, can now be expressed as structured markdown instructions that an AI agent executes autonomously. The agent reads the checklist, runs the specified checks, collects all findings, and presents them for human review—combining the thoroughness of automated scanning with the judgment-level assessments previously exclusive to human reviewers.

Example: a production checklist for book chapters. This book uses a markdown-based production checklist that an AI coding agent runs before each chapter is finalized. A condensed excerpt:

```
# Production Checklist for Chapter Review
The agent should: 1. Detect all problems first.
2. Present a PASS/FAIL table. 3. Ask the user
which to fix. 4. Apply only selected fixes.

## Checks to Run
### Cross-References
- All figures/tables/geek boxes referenced in text.
### Accessibility
- All figures have descriptive alt-text.
### Layout
- No "Float too large" or overfull hbox warnings.
### Acronyms
- No duplicate acronym expansions.
```

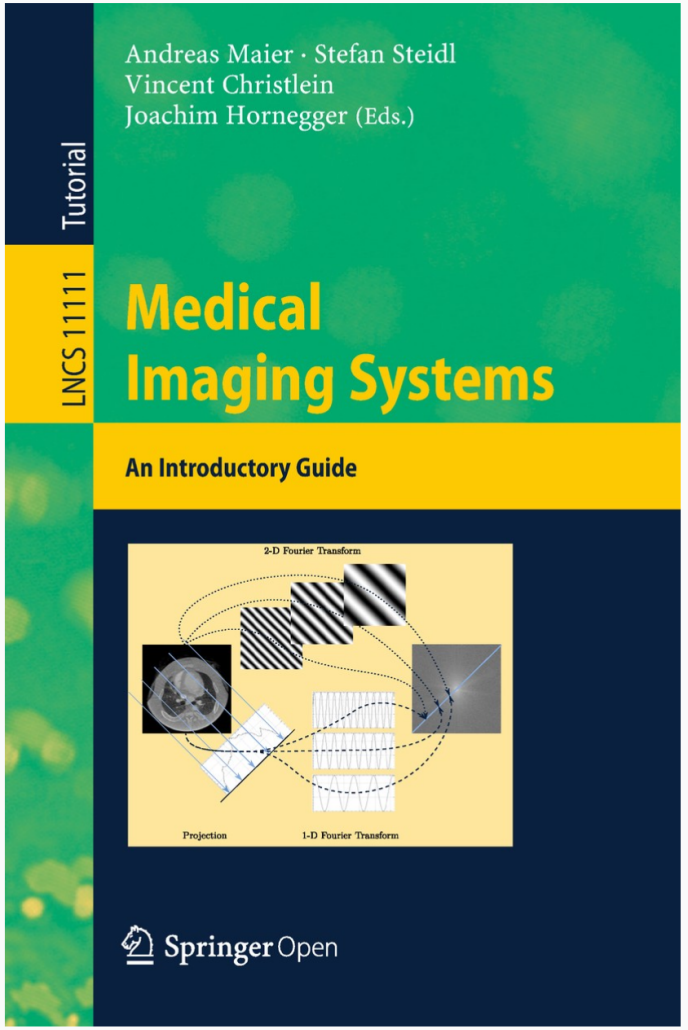
Several of these checks—alt-text quality, duplicate acronym detection, caption descriptiveness—require understanding the *meaning* of text, not just its syntax. An AI agent can evaluate whether alt-text actually describes what a figure shows or whether a manually spelled-out acronym conflicts with a `\gls{}` expansion nearby. These are assessments no simple script can perform reliably.

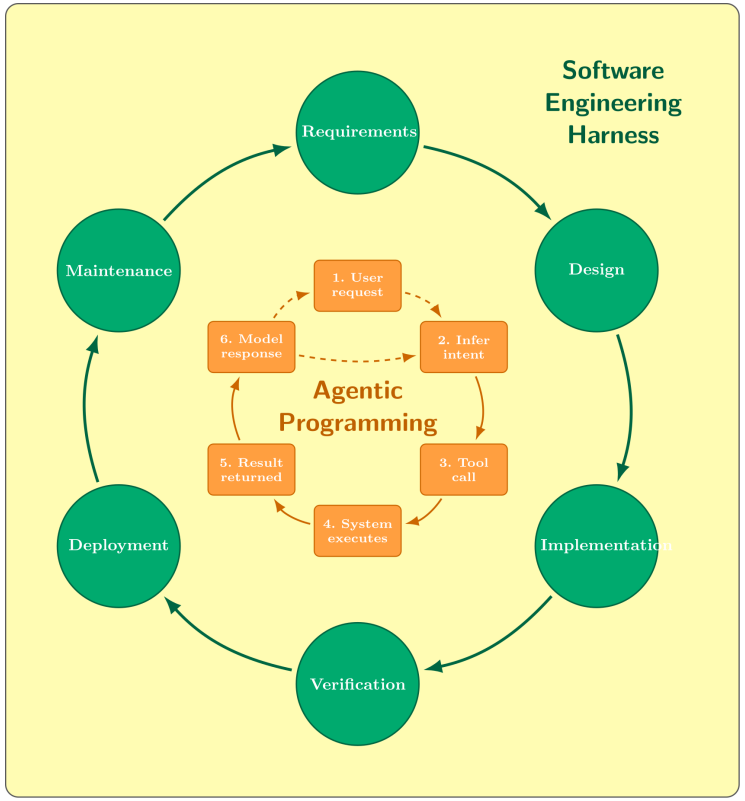
The detect-then-ask workflow. The checklist specifies a four-step process: detect all issues, present a PASS/FAIL summary, ask the user which fixes to apply, and then execute only approved changes. This separation of detection from correction mirrors the role-separation principle from Chapter 12: the agent that finds problems should not unilaterally fix them. The human decides which findings are genuine issues and which are acceptable.

This pattern generalizes beyond book production. Any domain with explicit quality criteria—code review checklists, regulatory compliance, accessibility audits, documentation reviews—can be encoded as markdown and executed by an agent. The quality gate becomes a reusable, version-controlled artifact rather than tacit knowledge in a reviewer's head.

What is agentic AI?

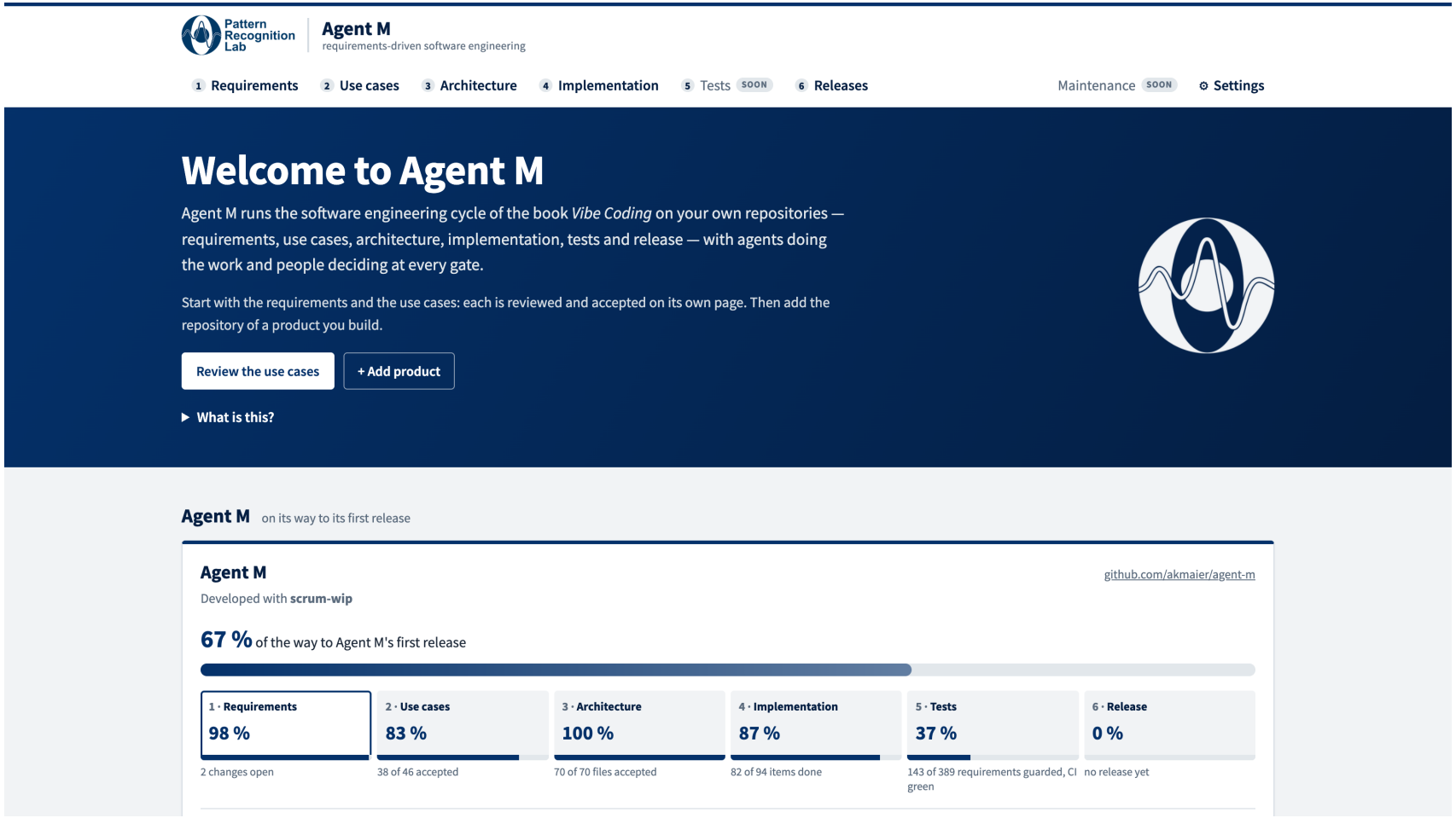
Textbook





What is agentic AI?

Agent M – the companion tool to the book *Vibe Coding*



akmaier.github.io/agent-m

What does it mean for science?

Almost everything is easy now...

China and AI



Almost everything is easy now...

Conference shares

Conference	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
CVPR	~30%	32%	35%	35%	39%	39.2%	~40%	~38%	~39%	~40%
ICCV/ECCV	~30%	30%	32%	35%	~37%	~40%	~40%	~38%	~40%	~40%
ICML	~10%	12%	15%	18%	~20%	~22%	~25%	~28%	~30%	~30%
NeurIPS	8%	10%	12%	15%	*~15%*	~18%	~20%	~25%	~28%	~30%
AAAI	15%	18%	22%	28%	~35%	50%	~50%	~45%	~48%	~50%

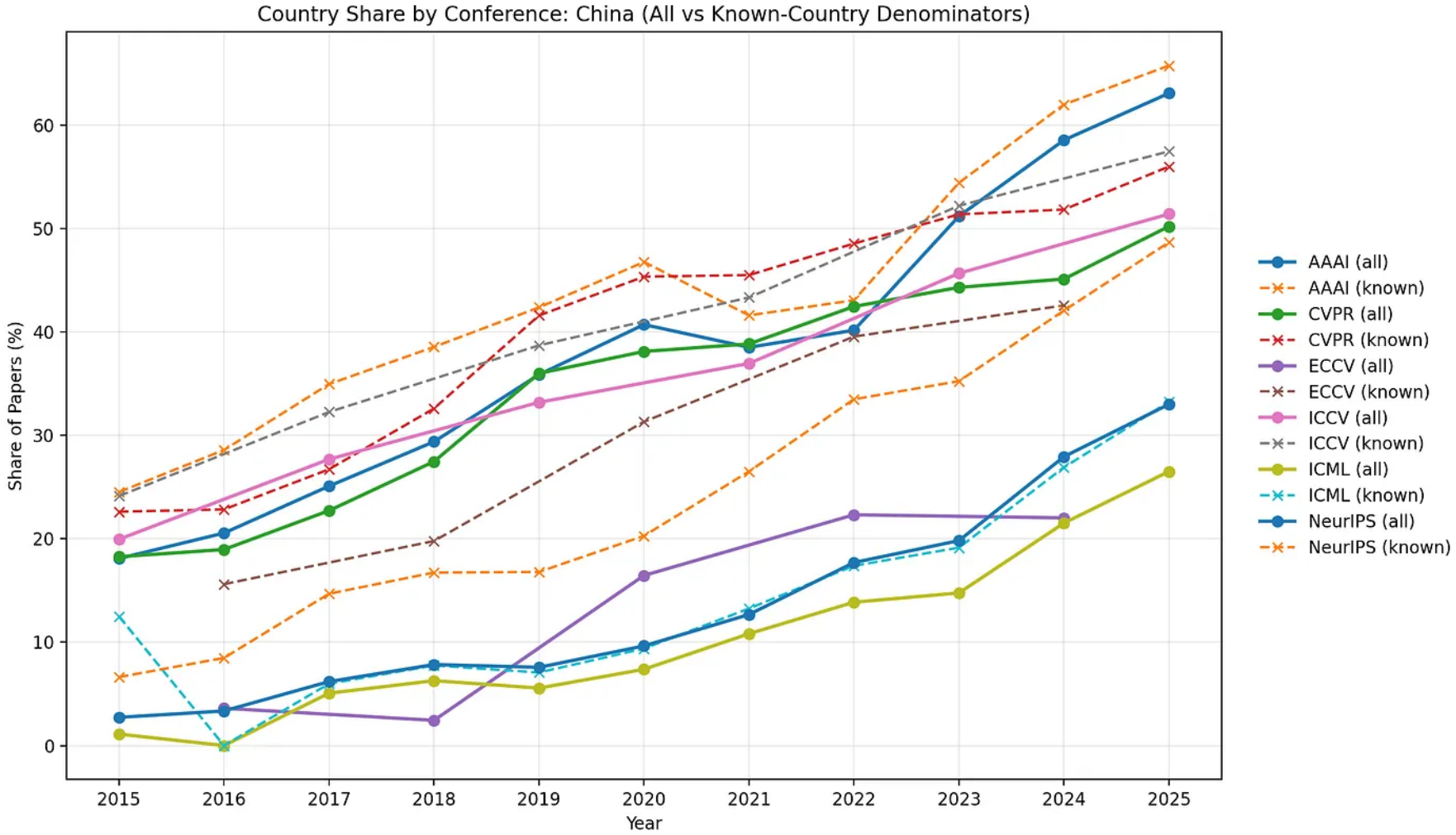
Almost everything is easy now. . .

ConferenceStats workflow



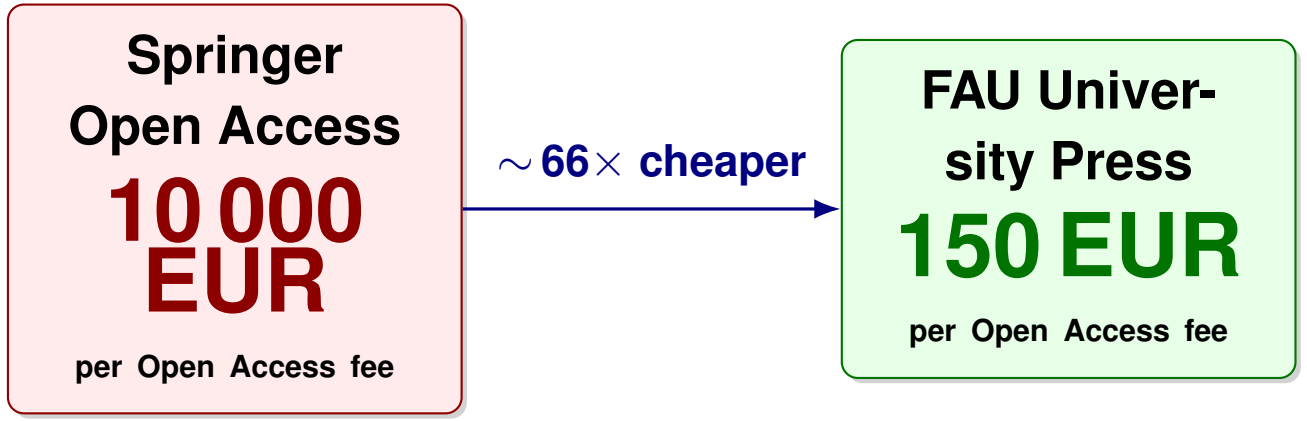
Almost everything is easy now…

Conference contributions from China



Publishing, Disrupted

Open Access for the price of a pizza party



- **106** submissions → **93** accepted → one Open Access volume.
- Idea to printed book: **three months**, including **peer review**.

Agent4CT

All deep-learning CT methods, in one weekend

- Continuously-running LLM agent that edits a CT reconstruction codebase, runs short GPU experiments, and keeps changes only when the metrics agree. Five agents run in parallel across five CT challenges and pool findings through a shared scratch pad.

Re-implemented solvers (`pentathlon/demo_dl_reference`):

- **Analytical:** PyroNN FBP (Syben 2019).
- **Classical iterative:** TV (iterative, supervised, hyper-search); Wu *et al.* 2015 (+ trainable port).
- **Variational / unrolled:** Hammernik 2017 variational network (+ CT port); Learned Primal-Dual (Adler & Öktem); ITNet v1–v3.
- **Self-supervised denoising:** dual-domain U-Net and trainable bilateral filters (Wagner 2023 / 2022, with supervised variants).
- **Per-scene neural fields:** NAF (neural attenuation field); R2Gaussian.
- **Foundation models:** RAM (Terris 2025).
- **Diffusion priors:** DDPM; diffusion-prior sampling; diffusion-recon with data consistency.
- **Transformer:** U-Swin.

`github.com/akmaier/Agent4CT`

UnderstandingVCS

A differentiable Atari 2600 – built by agents, proven bit-exact

- **Two differentiable ports** of the Atari 2600 VCS – in JAX and in Julia.
- Checked against **xitari**, an independent C++ emulator that is not part of the project.
- **Ground truth for XAI**: score interpretability methods against a machine we know *exactly*.

"Hallucinated code does not accidentally match an external reference to the bit."

64 / 64
games: RAM byte-exact, screen pixel-exact

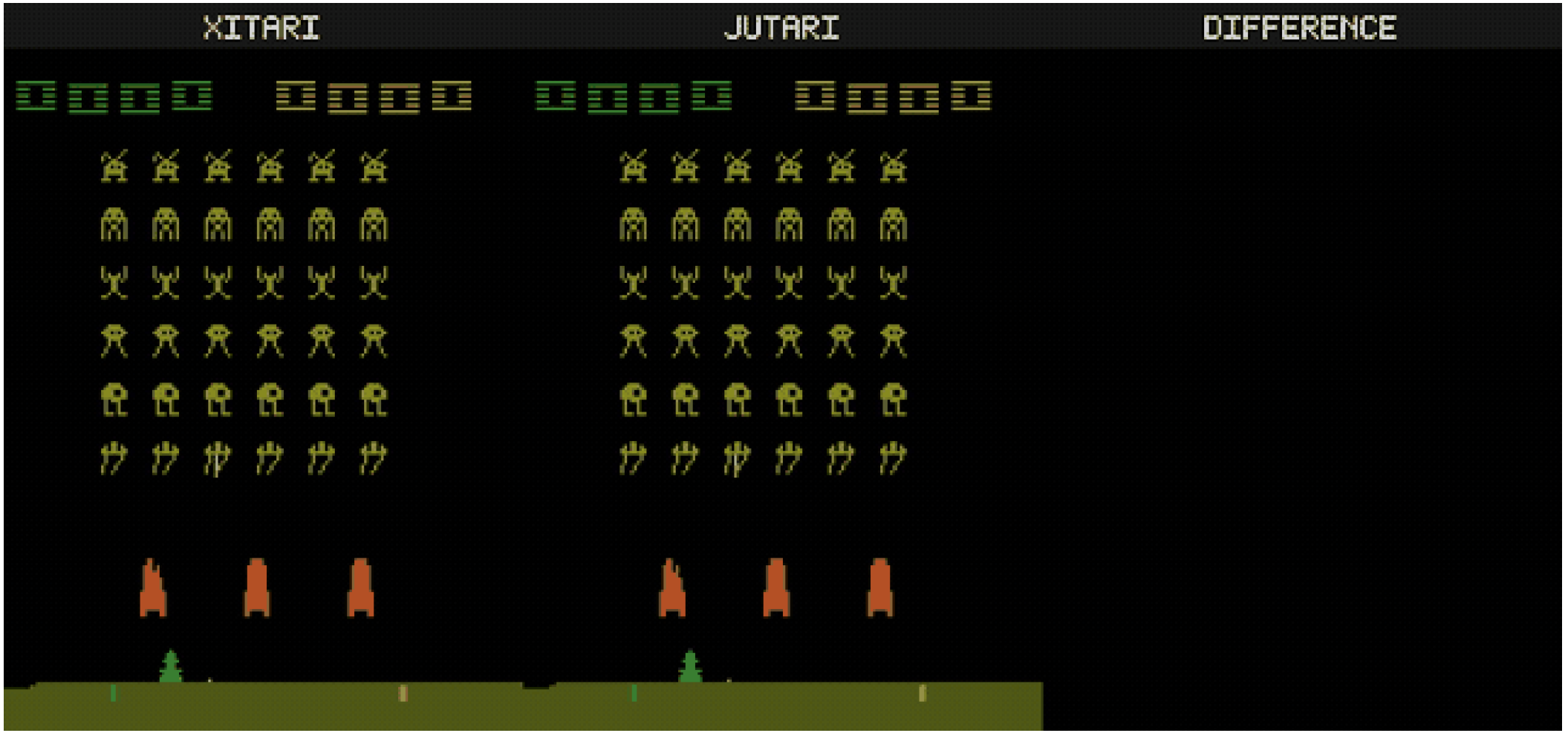
16 k
lines of code in one week

769
commits with the open prompt → code log

Paper 1: AAAI 2027 (submitted) • Paper 2: *Nature Machine Intelligence* (in preparation) • `akmaier.github.io/UnderstandingVCS`

UnderstandingVCS

Space Invaders – C++ reference vs. our Julia port: zero difference



Speed of Agentic AI

Wall-clock from idea to working artefact

3 h Reproduce a paper + code, find new insights, draft a follow-up paper.
`arxiv.org/abs/2605.02620`

1 d Reproduce a paper *without* its code or data.
`arxiv.org/abs/2605.12772`

3 d Scrape 200 study programs + ship a RAG chatbot.
`github.com/akmaier/IfCampoKnew`

1 w Reimplement an entire ATARI VCS emulator – bit-exact, 16 k lines of code.
`github.com/akmaier/UnderstandingVCS`

Science at Agent Speed

Opportunities and risks

Opportunities

- Months of work in hours.
- Reproduce papers without code.
- Open Access for 150 EUR.
- Coding for every discipline.

Risks

- Plausible is not correct.
- Hallucinated references.
- AI-text detectors beaten in 3 h.
- Sponsored bias in 12 LLMs.

Maier et al., arXiv:2605.02620 (style detector) • Maier et al., arXiv:2605.12772 (sponsored recommendations)

What This Means for Your PhD

Five rules for working with agents

1. Be the verifier, not the typist.
2. Reproduce before you build.
3. Keep a git-backed lab journal.
4. Write quality gates, not just prompts.
5. Disclose AI use – own the final word.

References

- V. Mnih et al., *Playing Atari with Deep Reinforcement Learning*, arXiv:1312.5602, 2013.
- D. Silver et al., *Mastering the game of Go with deep neural networks and tree search*, *Nature*, 529:484–489, 2016.
- **A. Maier** et al., *Learning with Known Operators Reduces Maximum Error Bounds*, *Nat. Mach. Intell.*, 1:373–380, 2019.
- M. Zaiss, J. R. Rajput, H. N. Dang, V. Golkov, D. Cremers, F. Knoll and **A. Maier**, *Exploring GPT-4 as MR sequence and reconstruction programming assistant: GPT4MR*, BVM Workshop, pp. 94–99, Springer, 2024.
- M. Zaiss, A. Aly, J. Endres, T. Dornstetter, S. Weinmüller and **A. Maier**, *Agentic MR sequence development: leveraging LLMs with MR skills for automatic physics-informed sequence development*, arXiv:2604.13282, 2026 (Agent4MR).
- Deutscher Zukunftspreis (low-field MRI): https://www.deutscher-zukunftspreis.de
- Agent M – companion tool to the book *Vibe Coding*: https://github.com/akmaier/agent-m
- Conference statistics tooling: https://github.com/akmaier/ConferenceStats
- Agent4CT – agentic re-implementation of deep-learning CT methods: https://github.com/akmaier/Agent4CT
- **A. Maier**, M. Zaiss and S. Bayer, *Beating the Style Detector: Three Hours of Agentic Research on the AI-Text Arms Race*, arXiv:2605.02620, 2026.
- **A. Maier**, J. Sopa, G. G. Sahin, P. Perez-Toro and S. Bayer, *Just Ask for a Table: A Thirty-Token User Prompt Defeats Sponsored Recommendations in Twelve LLMs*, arXiv:2605.12772, 2026.
- *IfCampoKnew* – scraper for 200 study programs and an AI RAG chatbot: https://github.com/akmaier/IfCampoKnew
- *UnderstandingVCS* – agent-reimplemented ATARI VCS emulator: https://github.com/akmaier/UnderstandingVCS

Thank you! Questions?

Slides on GitHub



`github.com/akmaier/AIMed2026Slides/tree/puls-retreat-2026`

Hands-on AI today, 16:30–18:30 – bring your own problem!